Technical Appendix

Supporting Documentation for the Causal and Predictive Marketing Analytics Report

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Table of Contents **2**
A. DATA DICTIONARY **3**
A.1 Raw Tables 3
A.2 Engineered Features - Clustering 3
A.3 Engineered Features - RFM 3
A.4 Engineered Features - Conjoint 4
A.5 Engineered Features - Logistic Regression 4
A.6 Engineered Features - Forecasting 4
B. FEATURE ENGINEERING LOGIC **4**
B.1 Data Merging Pipeline 5
B.2 Cluster Feature Selection 5
B.3 RFM Scoring 5
B.4 Conjoint Variable Construction 5
B.5 Logistic Regression Target Construction 5
C. DIAGNOSTICS AND VALIDATION **7**
C.1 Clustering Diagnostics 7
C.2 RFM Validation 7
C.3 Conjoint Model Diagnostics 7
C.4 Logistic Regression Diagnostics 7
C.5 Forecasting Validation 8
D. PARAMETER CHOICES AND SENSITIVITY **9**
E. ALTERNATIVE SPECIFICATIONS CONSIDERED **10**
E.1 Clustering Alternatives 10
E.2 RFM Alternatives 10
E.3 Conjoint Alternatives 10
E.4 Logistic Regression Alternatives 10
E.5 Forecasting Alternatives 10
F. SOFTWARE AND REPRODUCIBILITY **11**
F.1 Environment 11
F.2 File Structure 11
F.3 Reproduction Instructions 11

A. DATA DICTIONARY

This section documents all variables used across the five analyses. The raw dataset consists of five relational tables sourced from Kaggle.

A.1 Raw Tables

Table	Rows	Primary Key	Description
sales	1,000,000+	sale_id	Individual transaction records with store, product, date, quantity
products	~150	product_id	Product catalogue with name, category, price, launch date
category	10	category_id	Product category lookup (Smartphone, Laptop, Audio, etc.)
stores	50	store_id	Store information with name, city, country
warranty	~50,000	claim_id	Warranty claim records linked to sales with repair status and date

A.2 Engineered Features - Clustering

All clustering features were aggregated at the **store_id × category_name** level (750 observations).

Variable	Definition	Derivation
total_revenue	Total revenue per store-category	SUM(price × quantity)
total_units_sold	Total units sold	SUM(quantity)
avg_order_value	Average order value	total_revenue / num_transactions
avg_price_point	Average product price	MEAN(price)
num_distinct_products	Number of unique products sold	NUNIQUE(product_id)
total_claims	Total warranty claims filed	COUNT(claim_id) where not null
rejection_rate	Proportion of claims rejected	MEAN(repair_status == 'Rejected')
completion_rate	Proportion of claims completed	MEAN(repair_status == 'Completed')
avg_claim_latency_days	Mean days between sale and claim	MEAN(claim_date - sale_date)
claim_rate	Claims per unit sold	total_claims / total_units_sold

A.3 Engineered Features - RFM

RFM metrics were computed at the **store_id × category_name** level, consistent with the clustering unit.

Variable	Definition	Scoring
recency	Days since last transaction (relative to reference date)	Quintile-based, 1–5 (lower recency = higher score)
frequency	Number of unique transactions	Quintile-based, 1–5 (higher frequency = higher score)
monetary	Total revenue generated	Quintile-based, 1–5 (higher monetary = higher score)
RFM_score	Combined RFM score	R_score + F_score + M_score (range: 3–15)
segment	Behavioural segment label	Rule-based assignment (see segmentation rules below)

A.4 Engineered Features - Conjoint

Variable	Definition	Derivation
sub_brand	Product sub-brand (e.g., MacBook Pro, iPhone)	Extracted from product_name, splitting before parenthetical
product_age_days	Days between product launch and sale	sale_date - launch_date
price_tier	Price quartile bucket	pd.qcut(price, q=4) → Low, Mid, High, Premium
ln_quantity	Log-transformed total quantity	ln(total_quantity + 1)

A.5 Engineered Features - Logistic Regression

Features aggregated at the **store_id × category_name × sale_month** level.

Variable	Definition	Derivation
high_claim_risk (target)	Binary: 1 if claim_rate in top 25%	claim_rate > 75th percentile of monthly claim rates
total_quantity	Total units sold in the month	SUM(quantity)
avg_price	Average price in the month	MEAN(price)
total_revenue	Monthly revenue	SUM(quantity × price)
avg_product_age_days	Mean product age at sale	MEAN(sale_date - launch_date)

A.6 Engineered Features - Forecasting

Variable	Definition	Derivation
sale_month	Monthly timestamp	sale_date truncated to month start
monthly_revenue	Total revenue per cluster-category-month	SUM(quantity × price) grouped by cluster_name, category_name, sale_month

B. FEATURE ENGINEERING LOGIC

B.1 Data Merging Pipeline

The five raw tables were merged sequentially using left joins, with row-count assertions after each join to detect fan-out. The merge order was: sales → products (on product_id) → category (on category_id) → stores (on store_id) → warranty (on sale_id). The warranty join introduced a slight row increase due to multiple claims per sale in some cases. The merged dataset was exported as apple_merged.csv.

Column names were standardised to lowercase with underscores. Date columns (sale_date, launch_date, claim_date) were parsed to datetime format. A derived revenue column (price × quantity) was created for all downstream analyses.

B.2 Cluster Feature Selection

An initial set of 10 features was proposed. A correlation audit identified highly correlated pairs ($|r| > 0.85$), and redundant features were dropped to avoid multicollinearity in the distance-based clustering. Specifically, total_units_sold was highly correlated with total_revenue and num_transactions. After dropping redundant features, the final feature set contained 7–8 variables. All features were z-score standardised (mean=0, std=1) using sklearn StandardScaler.

B.3 RFM Scoring

Quintile-based scoring was used rather than fixed thresholds to ensure equal bin sizes (20% of observations per score level). The rank(method='first') approach was used for recency to handle ties deterministically. Segment assignment used a rule-based function with the following logic:

Segment	Rule
Champions	R >= 4 AND F >= 4 AND M >= 4
Loyal	R >= 3 AND F >= 3 AND M >= 3
New/Promising	R >= 4 AND F <= 2
At Risk	R <= 2 AND F >= 3
Hibernating	R <= 2 AND F <= 2 AND M <= 2
Need Attention	All others

B.4 Conjoint Variable Construction

Sub-brands were extracted from product names by splitting before parenthetical details and stripping inch specifications. Rare sub-brands (fewer than 50 observations) were collapsed into an “Other” category to prevent overfitting. The dependent variable was log-transformed (ln(quantity + 1)) to approximate normality and allow coefficient interpretation as approximate percentage changes. Data was aggregated to the product × store level, with total quantity sold as the dependent variable.

B.5 Logistic Regression Target Construction

The binary target variable (high_claim_risk) was defined as the top 25th percentile of monthly claim rates at the store-category-month level. This threshold was chosen to create a balanced-enough classification problem while focusing on genuinely elevated risk. The 75th percentile threshold was computed from the entire dataset before train/test splitting to ensure consistent labelling.

C. DIAGNOSTICS AND VALIDATION

C.1 Clustering Diagnostics

Optimal k selection: The Elbow Method (plotting inertia vs. k for k=2 to k=8) showed a clear bend at k=3. Silhouette Analysis confirmed k=3 as the highest silhouette score. Both methods independently supported the same choice.

Cluster stability: KMeans was run with n_init=10 (10 different random initialisations) and random_state=42 for reproducibility. The cluster distribution was 104 / 207 / 439, showing reasonable balance with no singleton or near-empty clusters.

PCA visualisation: A 2-component PCA projection was used to visually confirm cluster separation. The first two principal components captured sufficient variance to show distinct cluster boundaries, particularly for the High-Rejection Outlier cluster.

Feature variance check: Coefficient of variation (CV) was computed for each feature. Features with near-zero variance were flagged. High-correlation pairs were identified and redundant features removed before clustering.

C.2 RFM Validation

Score distribution: Histogram of the composite RFM score (3–15 range) was inspected for uniformity. The distribution showed reasonable spread with no extreme concentration at boundary values.

Segment coherence: Each segment was profiled on revenue, frequency, recency, claim rate, and geographic distribution. Profiles were consistent with expected segment definitions (e.g., Champions had the highest monetary values; Hibernating had the lowest frequency).

Revenue-risk matrix: A scatter plot of monetary value versus claim rate, coloured by segment, was used to validate that segment assignments aligned with business intuition.

C.3 Conjoint Model Diagnostics

Model specification: Base model: $\ln(\text{quantity}) = \beta 0 + \beta(\text{category}) + \beta(\text{sub_brand}) + \beta(\text{price}) + \beta(\text{product_age}) + \varepsilon$. Interaction model added category × price terms.

Robust standard errors: HC3 heteroscedasticity-consistent standard errors were used throughout to account for non-constant variance. F-statistics were significant ($p < 0.001$) for both models, confirming joint significance of predictors.

Low R-squared discussion: The base model R-squared was approximately 0.01. This is expected given: (a) the synthetic nature of the dataset, which introduces more noise than real transactional data; (b) the product-store level aggregation, which averages out much of the meaningful variation; (c) the absence of true customer preference variation. The model's value lies in the relative ranking of part-worths and price sensitivities, not in point prediction accuracy.

Alternative specification: Cluster-level sub-models were estimated to test whether preferences vary by segment. While coefficient magnitudes differed across clusters, the small subsample sizes in Clusters 0 and 1 reduced precision.

C.4 Logistic Regression Diagnostics

Train/test split: 70/30 stratified split. The pipeline handled missing values (SimpleImputer, median for numeric, most frequent for categorical), one-hot encoding for categorical variables, and standard scaling for numeric features.

ROC-AUC: The model achieved AUC ≈ 0.57 on the test set. While modest, this is above the 0.50 random baseline and indicates the model captures some signal. A confusion matrix and classification report were generated to assess precision, recall, and F1-score across both classes.

Coefficient interpretation: The top positive coefficients (indicating higher claim risk) included subscription service category, Core Hardware Revenue Drivers cluster, and higher transaction quantities. Top negative coefficients (lower claim risk) included Smart Speaker and Streaming Device categories.

Limitations acknowledged: The moderate AUC suggests that warranty claim risk is driven by factors not captured in the dataset (e.g., specific product defects, customer usage patterns, regional service quality). The model should be treated as a directional indicator rather than a precision classifier.

C.5 Forecasting Validation

Model choice: Holt-Winters Exponential Smoothing with additive trend and additive seasonality was selected. This method is appropriate for monthly time series with potential trend and seasonal patterns.

Evaluation metrics: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were computed on historical data for each cluster-category series. The best-performing series (Accessories, Audio, Laptop in Core Hardware) had forecast-to-historical ratios close to 1.0, indicating stable extrapolation.

Forecast anomaly: The Smartphone series in the Core Hardware cluster produced a negative forecasted revenue value in one period. This is a known artefact of the Holt-Winters method when the last observed data point is unusually low, causing the additive trend component to extrapolate below zero. This should be clipped to zero or investigated with additional data before use in operational planning.

D. PARAMETER CHOICES AND SENSITIVITY

Analysis	Parameter	Value Chosen	Justification
Clustering	Number of clusters (k)	3	Elbow method + silhouette score both support k=3
Clustering	Scaling method	Z-score (StandardScaler)	Required for K-Means distance calculations; features on different scales
Clustering	Correlation threshold for feature removal	$\|r\| > 0.85$	Standard threshold to remove redundant features
Clustering	n_init (KMeans restarts)	10	Default; sufficient for dataset size
Clustering	random_state	42	Fixed seed for reproducibility
RFM	Scoring method	Quintile-based (1–5)	Ensures equal bin sizes; robust to outliers
RFM	Reference date	Last sale date + 1 day	Standard practice; ensures all transactions have recency >= 1
Conjoint	Dependent variable	ln(total_quantity + 1)	Log transform for approximate normality and % interpretation
Conjoint	Standard errors	HC3 (robust)	Accounts for heteroscedasticity without assuming constant variance
Conjoint	Rare sub-brand threshold	< 50 obs → Other	Prevents overfitting from sparse dummy variables
Logistic Reg.	Target definition	Top 25% claim rate = high risk	Balances class distribution while focusing on elevated risk
Logistic Reg.	Train/test split	70/30	Standard split; sufficient test set for evaluation
Logistic Reg.	Regularisation	Default (C=1.0, L2 penalty)	sklearn default; no hyperparameter tuning performed
Forecasting	Method	Holt-Winters Exponential Smoothing	Handles trend + seasonality in monthly series
Forecasting	Seasonality	Additive	Appropriate when seasonal variation is roughly constant in absolute terms
Forecasting	Forecast horizon	6 months	Reasonable planning horizon; beyond this, uncertainty increases substantially

E. ALTERNATIVE SPECIFICATIONS CONSIDERED

E.1 Clustering Alternatives

k=4 and k=5: Higher cluster counts were tested but resulted in clusters that were not meaningfully distinct in terms of business profiles. The additional clusters split the Core Hardware segment without adding actionable differentiation. k=3 provided the most interpretable and actionable segmentation.

DBSCAN: Density-based clustering was considered but not pursued because the feature space had relatively uniform density after scaling, and DBSCAN's performance is sensitive to the eps parameter which is difficult to set objectively in high-dimensional space.

E.2 RFM Alternatives

Store-level RFM: Initially tested but abandoned because all 50 stores showed highly uniform RFM profiles (all stores are active daily with similar revenue levels). Pivoting to store-category level introduced meaningful variation across the three RFM dimensions.

Fixed-threshold scoring: Considered using fixed percentile thresholds (e.g., top 20% = 5) instead of quintile-based scoring. Quintile-based was preferred because it guarantees equal group sizes and is more robust to the specific distribution shape.

E.3 Conjoint Alternatives

Choice-based conjoint (survey): Not feasible because we lacked survey data. The revealed-preference approach using observed purchase data was the best available alternative.

Log-log specification: A log-log model (ln(quantity) ~ ln(price)) was considered to directly estimate price elasticities. However, the semi-log specification was retained because it allows coefficient interpretation as absolute (per-dollar) price effects, which is more intuitive for managerial communication.

E.4 Logistic Regression Alternatives

Random Forest classifier: Considered as an alternative to logistic regression. While it might achieve higher AUC, the interpretability advantage of logistic regression (clear coefficient signs and magnitudes) was prioritised given the report's managerial audience.

Different risk threshold: The 75th percentile threshold for defining "high risk" was tested against 80th and 90th percentiles. The 75th percentile provided the best balance between class representation and focus on genuinely elevated risk.

E.5 Forecasting Alternatives

ARIMA: Considered but Holt-Winters was preferred for its simplicity and built-in handling of trend and seasonality without the need for differencing or manual order selection (p, d, q).

Prophet: Facebook Prophet was considered but adds unnecessary complexity for this use case. Holt-Winters provides comparable performance on short monthly series with clear trend patterns.

F. SOFTWARE AND REPRODUCIBILITY

F.1 Environment

Language: Python 3.x

Tool/Platform: Upload the .ipynb file on Google collab, upload all the datasets listed below and run all cells

Key libraries: pandas, numpy, scikit-learn (KMeans, StandardScaler, PCA, LogisticRegression, train_test_split, classification_report, confusion_matrix, roc_auc_score), statsmodels (OLS, robust standard errors), matplotlib, seaborn, scipy.

Forecasting: statsmodels.tsa.holtwinters (ExponentialSmoothing).

F.2 File Structure

File	Description
Big_Data_Analytics_Group_Project.ipynb	Main Jupyter notebook containing all code for the five analyses, reproducible end-to-end
category.csv, products.csv, sales.csv, stores.csv, warranty.csv	Raw data files (sourced from Kaggle)
apple_merged.csv	Merged dataset (output of data preparation pipeline)
apple_merged_clustered.csv	Merged dataset with cluster labels appended
store_category_clusters.csv	Store-category cluster assignments (output of clustering)
Marketing_Analytics_Report.docx	Main marketing analytics report (Deliverable I)
Technical_Appendix.docx	This document (Deliverable II)

F.3 Reproduction Instructions

1. Place all raw CSV files (category.csv, products.csv, sales.csv, stores.csv, warranty.csv) in the same directory as the notebook.

2. Open Big_Data_Analytics_Group_Project.ipynb in Jupyter Notebook or JupyterLab.

3. Run all cells sequentially from top to bottom. The notebook will: (a) load and merge the raw data, (b) perform clustering, (c) run RFM analysis, (d) execute conjoint analysis, (e) fit logistic regression, and (f) generate forecasts.

4. Intermediate CSV files (apple_merged.csv, apple_merged_clustered.csv, store_category_clusters.csv) will be created automatically during execution.

5. All figures and tables shown in the report can be reproduced by running the corresponding notebook cells.